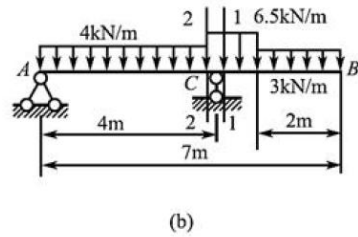
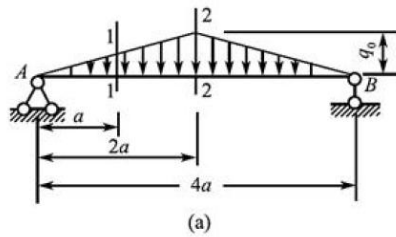


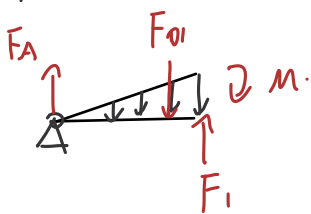
4-1 试求图4-2-1所示各梁中指定截面上的剪力和弯矩。



解: 由对称性 $F_A = F_B = \frac{1}{2} \cdot 2a \cdot \frac{q_0}{2} = a \cdot \frac{q_0}{2}$ 图4-2-1

(a) 1-1:

截面法取1-1面左边部分:



$$\sum F_y = 0 \Rightarrow F_A + F_1 = F_{01} = \frac{1}{2} \cdot a \cdot \frac{q_0}{2} = \frac{1}{4} a q_0$$

$$\therefore F_1 = \frac{1}{4} a q_0 - a q_0 = -\frac{3}{4} a q_0$$

$$\sum M_A = 0 \Rightarrow M_1 + \frac{1}{4} a q_0 \cdot \frac{2}{3} a = F_1 \cdot a$$

$$\therefore M_1 = -\frac{11}{12} a^2 q_0$$

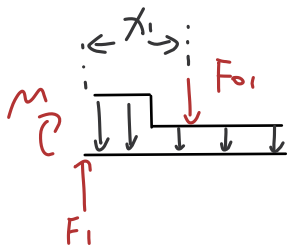
(a) 2-2:

由对称性. 将求解1-1答案中的 a 替换为 $2a$.

$$\therefore F_2 = -\frac{3}{2} a q_0$$

$$M_2 = -\frac{11}{3} a^2 q_0$$

(b) 截面法取1-1右边部分:



$$F_{01} = [(7-4-2) \times 6.5 + 2 \times 3] = -12.5 \text{ kN}$$

$$x_1 = \frac{\int x q(x) dx}{F_{01}} = \frac{\frac{1}{2} \times 6.5 \times 2^2 + \frac{3}{2} (2^2 - 1^2)}{12.5} = \frac{30.5}{12.5} = 2.5 \text{ m}$$

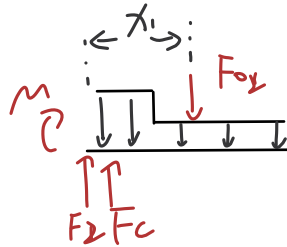
$$\sum F_y = 0 \Rightarrow F_{01} + F_1 = 0$$

$$F_1 = 12.5 \text{ kN}$$

$$\sum M = 0 \Rightarrow M + F_{01} \cdot x_1 = 0$$

$$\therefore M = 15.25 \text{ kN} \cdot \text{m}$$

取 2-2 右边部分



$$F_0 = (4 \times 4 + 1 \times 6.5 + 2 \times 3) \text{ N} = 28.5 \text{ N}$$

$$x_0 = \frac{\int q(x) \cdot x dx}{F_0} = \frac{\frac{4}{2} \times 4^2 + \frac{6.5}{2} \times (5^2 - 4^2) + \frac{3}{2} \times (7^2 - 5^2)}{28.5} = \frac{389}{114} \text{ m}$$

$$\text{对整体} \begin{cases} \sum F_y = 0 & F_A + F_c = F_0 \\ \sum M_A = 0 & F_0 \cdot x_0 = F_c \cdot 4 \end{cases}$$

$$\text{对局部} \begin{cases} F_2 + F_c = F_{02} \\ x_1 F_{02} + M = 0 \end{cases} \quad \therefore \begin{aligned} F_2 &= -11.8125 \text{ kN} \\ M_2 &= -15.25 \text{ kN} \cdot \text{m} \end{aligned}$$